Math 54 — Week 2

Solutions

P1.

Idea. By a vector equation is a linear system (Lay 1.3), $\mathbf{b}$ is a combination exactly when the system with augmented matrix $[\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\ \mathbf{b}\,]$ is consistent, and its solution gives the weights.

1. Row reduce:
$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -1 & 1 \\ 2 & 3 & 4 & 12 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 2 & 2 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$
2. Back-substitute:
$$x_3 = 3, \qquad x_2 = 1 + x_3 = 4, \qquad x_1 = 5 - 2x_2 - x_3 = -6.$$
3. So $\mathbf{b} = -6\mathbf{a}_1 + 4\mathbf{a}_2 + 3\mathbf{a}_3$. Check:
$$-6\begin{bmatrix}1\\0\\2\end{bmatrix} + 4\begin{bmatrix}2\\1\\3\end{bmatrix} + 3\begin{bmatrix}1\\-1\\4\end{bmatrix} = \begin{bmatrix}-6+8+3\\0+4-3\\-12+12+12\end{bmatrix} = \begin{bmatrix}5\\1\\12\end{bmatrix}.$$

P2.

Idea. $\mathbf{y}$ is in the span exactly when $[\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{y}\,]$ is consistent; keep h as a letter.

1. Row reduce:
$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 5 & h \\ -1 & 0 & 1 \end{array}\right] \begin{array}{c} R_2-2R_1 \\ \longrightarrow \\ R_3+R_1 \end{array} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & h-6 \\ 0 & 2 & 4 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & h-6 \\ 0 & 0 & 16-2h \end{array}\right]$$
2. The last row is the equation $0 = 16 - 2h$, so the system is consistent exactly when $h = 8$.
3. For $h = 8$: $x_2 = h - 6 = 2$ and $x_1 = 3 - 2x_2 = -1$, so $\mathbf{y} = -\mathbf{v}_1 + 2\mathbf{v}_2$. Check: $(-1,-2,1) + (4,10,0) = (3,8,1)$.

P3.

Idea. The columns span $\mathbb{R}^3$ exactly when there is a pivot in every row (Theorem 4); for the set of good $\mathbf{b}$, reduce with $\mathbf{b}$ as letters.

1. For A:
$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 2 & 3 & 0 & b_2 \\ 1 & 3 & -3 & b_3 \end{array}\right] \begin{array}{c} R_2-2R_1 \\ \longrightarrow \\ R_3-R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 0 & -1 & 2 & b_2-2b_1 \\ 0 & 1 & -2 & b_3-b_1 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 0 & -1 & 2 & b_2-2b_1 \\ 0 & 0 & 0 & b_2+b_3-3b_1 \end{array}\right]$$
Row 3 of the echelon form of A has no pivot, so the columns of A do not span $\mathbb{R}^3$. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $-3b_1 + b_2 + b_3 = 0$, a plane through $\mathbf{0}$. Check: the columns satisfy it, since $-3+2+1=0$, $-6+3+3=0$, $3+0-3=0$.
2. For B: $R_3 - 2R_1$ turns row 3 into $(0,0,1)$, giving the echelon form
$$\begin{bmatrix}1&2&0\\0&1&3\\0&0&1\end{bmatrix}.$$
There is a pivot in every row, so the columns of B span $\mathbb{R}^3$.

P4.

Idea. The definition combines the columns with weights 5 and -2; the row-vector rule takes one sum of products per row.

1. By the definition:
$$5\begin{bmatrix}2\\0\\1\end{bmatrix} - 2\begin{bmatrix}-1\\3\\4\end{bmatrix} = \begin{bmatrix}10+2\\0-6\\5-8\end{bmatrix} = \begin{bmatrix}12\\-6\\-3\end{bmatrix}.$$

2. By the row-vector rule, the three entries are
$$2\cdot 5 + (-1)(-2) = 12, \qquad 0\cdot 5 + 3\cdot(-2) = -6, \qquad 1\cdot 5 + 4\cdot(-2) = -3.$$
The two methods agree.

P5.

Idea. Reduce A (the zero column stays zero), solve for the basic variables, and factor out each free variable.

1. Row reduce:
$$\begin{bmatrix}1&3&-3&7\\0&1&-4&5\\2&7&-10&19\end{bmatrix} \xrightarrow{R_3-2R_1} \begin{bmatrix}1&3&-3&7\\0&1&-4&5\\0&1&-4&5\end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix}1&3&-3&7\\0&1&-4&5\\0&0&0&0\end{bmatrix} \xrightarrow{R_1-3R_2} \begin{bmatrix}1&0&9&-8\\0&1&-4&5\\0&0&0&0\end{bmatrix}$$

2. x_3, x_4 are free, $x_1 = -9x_3 + 8x_4$ and $x_2 = 4x_3 - 5x_4$. So
$$\mathbf{x} = x_3\begin{bmatrix}-9\\4\\1\\0\end{bmatrix} + x_4\begin{bmatrix}8\\-5\\0\\1\end{bmatrix}.$$

3. Check with the row-vector rule:
$$A\begin{bmatrix}-9\\4\\1\\0\end{bmatrix} = \begin{bmatrix}-9+12-3\\4-4\\-18+28-10\end{bmatrix} = \mathbf{0}, \qquad A\begin{bmatrix}8\\-5\\0\\1\end{bmatrix} = \begin{bmatrix}8-15+7\\-5+5\\16-35+19\end{bmatrix} = \mathbf{0}.$$

P6.

Idea. Parametric vector form: a particular solution plus the solutions of the homogeneous system (Theorem 6).

1. Call the coefficient matrix C. Row reduce $[\,C \;\; \mathbf{b}\,]$ with $\mathbf{b} = (1,4,5)$:
$$\left[\begin{array}{ccc|c}1&2&-1&1\\2&5&1&4\\3&7&0&5\end{array}\right] \xrightarrow[R_3-3R_1]{R_2-2R_1} \left[\begin{array}{ccc|c}1&2&-1&1\\0&1&3&2\\0&1&3&2\end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{ccc|c}1&2&-1&1\\0&1&3&2\\0&0&0&0\end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ccc|c}1&0&-7&-3\\0&1&3&2\\0&0&0&0\end{array}\right]$$

2. x_3 is free, $x_1 = -3 + 7x_3$ and $x_2 = 2 - 3x_3$, so
$$\mathbf{x} = \begin{bmatrix}-3\\2\\0\end{bmatrix} + x_3\begin{bmatrix}7\\-3\\1\end{bmatrix}.$$

P6, continued.

3. This is the line through $(-3, 2, 0)$ parallel to $(7, -3, 1)$. The solution set of $C\mathbf{x} = \mathbf{0}$ is the parallel line $\mathrm{Span}\{(7, -3, 1)\}$ through $\mathbf{0}$.
4. Check with the row-vector rule:

$$C\begin{bmatrix}-3\\2\\0\end{bmatrix} = \begin{bmatrix}-3+4\\-6+10\\-9+14\end{bmatrix} = \begin{bmatrix}1\\4\\5\end{bmatrix}, \qquad C\begin{bmatrix}7\\-3\\1\end{bmatrix} = \begin{bmatrix}7-6-1\\14-15+1\\21-21\end{bmatrix} = \mathbf{0}.$$

P7.

Idea. Count pivots: each pivot uses one row and one column (W3).

1. (a) False. A 3×2 matrix has at most 2 pivots, so some row has no pivot, and by Theorem 4 the columns do not span $\mathbb{R}^3$.
2. (b) True. By Theorem 4 there is a pivot in every row, so there are 4 pivots. The matrix has 4 columns, each with at most one pivot, so every column has a pivot and there is no free variable. The homogeneous system is consistent, so its only solution is $\mathbf{0}$.
3. (c) True. There are at most 3 pivots among 5 columns, so at least 2 free variables. The homogeneous system is consistent, so it has infinitely many solutions, and in particular a nontrivial one.

P8.

Idea. One equation: x_1 is basic and x_2, x_3 are free.

1. Solve for x_1: $x_1 = 4 - 2x_2 + 3x_3$. So

$$\mathbf{x} = \begin{bmatrix}4\\0\\0\end{bmatrix} + x_2\begin{bmatrix}-2\\1\\0\end{bmatrix} + x_3\begin{bmatrix}3\\0\\1\end{bmatrix}.$$

2. The homogeneous equation has the solution set $\mathrm{Span}\{(-2, 1, 0), (3, 0, 1)\}$, a plane through $\mathbf{0}$ (the two vectors are not multiples of each other).
3. The given equation's solution set is that plane moved by $(4, 0, 0)$: a parallel plane that does not contain $\mathbf{0}$, because $0 + 0 - 0 \neq 4$.